

Income Classification Project Report

3. Modeling and Hyperparameter Tuning

I selected Logistic Regression as the primary classifier for this income prediction task. To optimize the model's performance and understand its behavior under different constraints, I experimented with the 'C' parameter (the inverse of regularization strength).

Experimentation Results: Performance Metrics

Hyperparameter	Class	Precision	Recall	F1-Score	Accuracy
C = 0.01 (Strong Reg)	0	0.86	0.94	0.90	83.84%
	1	0.72	0.52	0.60	
C = 1.0 (Default)	0	0.87	0.92	0.90	83.84%
	1	0.70	0.56	0.62	
C = 100 (Weak Reg)	0	0.87	0.92	0.90	83.82%
	1	0.69	0.56	0.62	

Confusion Matrix Analysis (Default Hyperparameter C = 1.0)

Actual amount	Estimated 1 (Yes)	Estimated 0 (No)	Total
Actual 1 (Yes)	2,165 (TP)	1,681 (FN)	3,846
Actual 0 (No)	950 (FP)	11,480 (TN)	12,430
Total	3,115	13,161	16,276

Detailed Analysis of Hyperparameters

- **Strong Regularization (C = 0.01):** By imposing a stricter penalty on coefficients, the model became more conservative. This led to the highest Recall for Class 0 (94%), ensuring high accuracy for low-income identification.

- **Default Setting (C = 1.0):** This provided a robust baseline. As shown in the matrix above, the model successfully identified 2,165 high-income cases while maintaining balanced precision and recall.
- **Weak Regularization (C = 100):** Reducing the regularization allowed the model to fit the data more aggressively, leading to a slight increase in True Positives (2,169) but also increasing False Positives (957).

4. Data Visualization & Analysis

Based on the feature coefficients from the Logistic Regression model, key predictors were identified:

- **Education-num:** Strong positive correlation; higher education significantly increases the probability of earning >50K.
- **Age:** Positive trend reflecting professional experience and career progression.
- **Capital-gain:** Critical numeric factor for distinguishing individuals in the high-income bracket.

5. Conclusion

The Logistic Regression model performed consistently with an accuracy of approximately 84%. The evaluation shows that using the **Default Hyperparameter (C = 1.0)** ensures a balanced trade-off between precision and recall, making it an effective choice for identifying high-income individuals within this dataset.